

Deep Learning-Based Illegal Road Crossing Detection in Dhaka Urban Traffic Surveillance

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Abstract

Pedestrian-vehicle conflicts are a major safety concern in dense, mixed-traffic cities such as Dhaka, Bangladesh, where limited crossing infrastructure makes pedestrian detection alone insufficient for identifying unsafe road entry. To address this gap, we collected a dataset from six locations across Dhaka and introduced two annotation schemes: a single-class scheme for general pedestrian detection and a spatially aware three-class scheme comprising *person_middle*, *person_side*, and *person_sidewalk*. Four YOLO variants (YOLOv8m, YOLOv8l, YOLO11m, and YOLO11l) were evaluated on independent test sets. YOLO11l achieved the best single-class performance, with a test mAP@0.5:0.95 of 0.6394, recall of 0.8117, and mAP@0.5 of 0.8419, while YOLOv8m achieved the highest precision of 0.7748. For the multi-class task, YOLOv8m achieved the highest recall of 0.8519 and mAP@0.5:0.95 of 0.5004, whereas YOLOv8l achieved the highest precision of 0.8761. These results indicate that larger models do not consistently provide better generalization for spatially structured pedestrian classification. Resolution experiments further identified 960×960 as the best overall operating point across both tasks, particularly for recall and mAP@0.5, while 1280×1280 produced a slightly higher mAP@0.5:0.95 for the multi-class task. Overall, the results demonstrate the value of spatially aware annotations and task-specific model selection for fixed-camera pedestrian monitoring in complex urban environments.

1. Introduction

Pedestrian safety remains a critical challenge in modern intelligent transportation systems (ITS), as pedestrians represent one of the most vulnerable groups of road users. According to global road safety reports, pedestrian-related crashes contribute significantly to traffic injuries and fatalities worldwide, particularly in rapidly developing urban regions where increasing vehicle density and insufficient pedestrian infrastructure create complex interactions between vehicles and pedestrians [16]. Unlike controlled transportation environments, many metropolitan areas experience heterogeneous traffic patterns, weak lane discipline, and irregular pedestrian movement, increasing the risk of pedestrian-vehicle conflicts [19].

Illegal road crossing, where pedestrians enter active traffic lanes outside designated crossing facilities, is one of the major factors contributing to pedestrian safety risks. Such behavior is influenced by several factors, including inadequate crossing facilities, long waiting times, traffic congestion, and individual decision-making processes [19, 21]. This problem is particularly challenging in densely populated developing cities such as Dhaka, Bangladesh, where mixed traffic conditions, limited pedestrian infrastructure, roadside activities, and high pedestrian density create unpredictable road interactions. Therefore, automated monitoring of illegal road crossing behavior has become an important requirement for improving urban traffic safety and supporting intelligent transportation management systems [4].

The rapid expansion of surveillance camera networks and advances in computer vision have

created new opportunities for automated traffic monitoring. Vision-based ITS applications have been widely explored for pedestrian detection, vehicle monitoring, traffic analysis, and safety assessment [4, 32]. In particular, deep learning-based object detection methods have significantly improved the ability to detect pedestrians under challenging conditions compared with traditional handcrafted feature-based approaches such as Histogram of Oriented Gradients (HOG) and deformable part models (DPM) [9]. Modern convolutional neural network (CNN)-based detectors, including Faster R-CNN and single-stage detectors, have demonstrated strong performance in real-world visual recognition tasks [27, 32].

Among deep learning-based detectors, the You Only Look Once (YOLO) family has gained significant attention because of its balance between detection accuracy and computational efficiency. Unlike two-stage detectors that separate region proposal and classification processes, YOLO performs object localization and classification in a single forward pass, enabling real-time inference suitable for surveillance applications [2, 26]. Recent improvements in YOLO architectures, including YOLOv8, have further enhanced detection performance through improved feature extraction and optimization strategies, making them suitable for real-time pedestrian monitoring systems [11].

However, illegal road crossing detection is considerably more challenging than conventional pedestrian detection. Existing object detectors primarily determine the presence and location of pedestrians but do not directly understand pedestrian-road interaction. A pedestrian detected near a roadway may represent a person walking on a sidewalk, waiting near a crossing area, or illegally entering active traffic lanes. Therefore, accurate illegal crossing detection requires contextual understanding of pedestrian position, road boundaries, and surrounding traffic environments rather than simple object localization [18, 21].

Despite substantial progress in pedestrian detection and behavior analysis, existing datasets and approaches have several limitations for illegal road crossing detection. Large-scale pedestrian datasets such as Caltech Pedestrian and Cityscapes have significantly advanced pedestrian detection research by providing diverse urban scenes and bounding-box annotations [5, 7]. However, these datasets mainly focus on pedestrian localization and detection rather than identifying whether pedestrians are violating road crossing regulations. Similarly, behavior-oriented datasets such as JAAD and related intention-prediction benchmarks provide valuable information about pedestrian intention and crossing behavior, but they are primarily designed for autonomous driving scenarios and future trajectory prediction rather than illegal crossing detection in dense urban surveillance environments [14, 21].

Furthermore, many existing approaches rely on trajectory analysis, manually defined rules, or additional road geometry information to identify pedestrian crossing behavior. While these methods can perform effectively in structured environments, their performance may degrade in complex urban scenarios where road boundaries are unclear, pedestrian movement is irregular, and traffic conditions are highly dynamic [14, 18]. Additionally, most publicly available datasets are collected from developed transportation environments and may not adequately represent the characteristics of densely populated developing cities where informal crossing patterns are common.

These limitations reveal an important research gap: the absence of dedicated datasets and deep learning frameworks specifically designed for illegal pedestrian road crossing detection under realistic urban conditions. A practical solution requires not only accurate pedestrian detection but also the ability to distinguish different pedestrian states based on their interaction with road regions. To address this gap, this study introduces a custom dataset collected from multiple urban locations in Dhaka, Bangladesh, and proposes a YOLOv8-based framework for detecting pedestrian road crossing status from surveillance images.

The main contributions of this study are summarized as follows:

- A custom illegal road crossing detection dataset is developed with annotated pedestrian categories representing different road interaction states, including illegal crossing, roadside walking, and sidewalk usage.
- A YOLO-based real-time detection framework is proposed to identify pedestrian crossing behavior from surveillance images.
- The proposed framework is evaluated under challenging urban conditions, including occlusion, illumination variation, and complex traffic scenarios, demonstrating its potential for intelligent transportation safety applications.

The proposed framework aims to bridge the gap between conventional pedestrian detection and pedestrian-road interaction analysis by integrating deep learning with a dataset specifically designed for illegal road crossing detection. Through real-world data collection and evaluation, this study provides a foundation for developing automated pedestrian safety monitoring systems suitable for complex urban environments.

2. Related Work

Research related to illegal road crossing detection lies at the intersection of pedestrian detection, pedestrian behavior understanding, and intelligent traffic surveillance. While significant progress has been achieved in these areas, existing methods and datasets are primarily designed for pedestrian localization, autonomous driving, or crossing intention prediction rather than identifying illegal crossing events from fixed surveillance cameras. This section reviews the most relevant developments and highlights the limitations that motivate our work.

2.1. Pedestrian Detection

Pedestrian detection has been a fundamental research problem in computer vision due to its importance in intelligent transportation systems and autonomous driving applications. Early pedestrian detectors relied on handcrafted features and traditional classifiers. Dalal and Triggs introduced Histogram of Oriented Gradients (HOG) with Support Vector Machines (SVM), which became one of the most influential approaches for human detection by exploiting local gradient information [6]. Later, deformable part models (DPM) improved robustness by representing objects using deformable components; however, these methods remained limited under complex conditions such as heavy occlusion, illumination variation, and large appearance changes [10].

The emergence of deep convolutional neural networks significantly improved pedestrian detection performance by automatically learning hierarchical visual representations. Two-stage detectors, particularly Faster R-CNN, introduced Region Proposal Networks (RPNs) to efficiently generate object proposals and achieved strong detection accuracy on benchmark datasets such as Caltech Pedestrian and CityPersons [8, 28, 31]. However, the proposal-generation stage increases computational cost, limiting their suitability for real-time large-scale surveillance applications.

Single-stage detectors addressed the speed limitation by performing localization and classification in a single network inference. The YOLO framework reformulated object detection as a regression problem, enabling real-time detection while maintaining competitive accuracy [25]. Subsequent improvements, including YOLOv3, YOLOv4, YOLOv7, and YOLOv8, introduced stronger feature extraction networks, improved feature aggregation, and optimized training strategies to achieve better speed-accuracy trade-offs [1, 11, 24, 29]. Due to its real-time capability and deployment efficiency, YOLO-based architectures have become widely adopted for surveillance-based pedestrian analysis.

However, pedestrian detectors only provide object localization and do not determine whether a pedestrian is performing a legal or illegal road interaction. In illegal crossing detection, recognizing

pedestrian position relative to road regions is as important as detecting the pedestrian itself. Therefore, conventional pedestrian detection methods alone are insufficient for traffic violation understanding.

2.2. Pedestrian Behavior and Crossing Intention Analysis

Beyond pedestrian localization, researchers have investigated pedestrian behavior analysis and crossing intention prediction to improve vehicle-pedestrian interaction. The JAAD dataset introduced by Rasouli et al. provided pedestrian bounding boxes, behavioral annotations, and contextual attributes for studying whether pedestrians intend to cross roads [22]. The dataset established an important benchmark for pedestrian crossing behavior analysis; however, it was collected from ego-vehicle viewpoints and focuses on predicting crossing intention rather than determining whether a crossing is illegal.

The PIE dataset further extended pedestrian behavior research by introducing a large-scale benchmark containing more than 300,000 annotated frames with pedestrian locations, actions, intentions, and trajectory information [23]. Several approaches based on recurrent neural networks, trajectory prediction, and contextual modeling have demonstrated that temporal motion patterns and scene context provide valuable information for predicting pedestrian actions [13, 20].

Although these studies provide important insights into pedestrian decision-making, their objectives differ from illegal road crossing detection. Existing behavior datasets generally consider whether pedestrians cross or intend to cross, but they do not provide annotations describing whether the crossing violates road regulations. Furthermore, most approaches rely on moving vehicle cameras and additional contextual information that are unavailable in fixed surveillance environments.

2.3. Traffic Surveillance and Road Violation Detection

Computer vision-based traffic surveillance has been widely investigated for intelligent transportation applications, including traffic monitoring, anomaly detection, and safety violation analysis. Buch et al. presented a comprehensive review of computer vision techniques for road scene analysis, highlighting the importance of automated surveillance for transportation management and safety applications [3]. Recent deep learning-based systems have further improved traffic monitoring by enabling automatic analysis of complex road environments [33].

Several studies have explored automated traffic violation detection using object detection and tracking techniques. These systems commonly identify violations such as illegal parking, lane violations, and unsafe vehicle behavior using predefined road rules and spatial constraints. Similarly, pedestrian violation detection approaches have attempted to identify jaywalking by combining pedestrian detectors with road region estimation or trajectory analysis [12, 15, 30]. Although these approaches demonstrate the feasibility of automated illegal crossing detection, most are evaluated in structured road environments with clear road boundaries and limited pedestrian density.

Dense urban environments introduce additional challenges because pedestrians may be partially occluded, roads may lack clear markings, and traffic interactions are often unstructured. These limitations are particularly relevant for South Asian cities such as Dhaka, where mixed traffic patterns and informal pedestrian movement are common. Therefore, methods developed for structured traffic environments cannot be directly applied without region-specific data and adaptation.

2.4. Existing Datasets and Research Gap

The availability of annotated datasets has significantly contributed to progress in pedestrian vision research. The Caltech Pedestrian dataset provided one of the earliest large-scale pedestrian benchmarks collected from urban driving videos [8]. CityPersons extended pedestrian detection research by providing high-quality annotations from the Cityscapes dataset, including challenging cases with occlusion and crowded scenes [31]. More recently, datasets such as JAAD and PIE introduced behavioral annotations for pedestrian intention prediction and trajectory analysis [22, 23].

Despite these contributions, existing datasets have significant limitations for illegal road crossing detection. Pedestrian detection datasets mainly provide bounding boxes without behavioral information, while behavior datasets focus on crossing intention from vehicle-mounted cameras rather than illegal crossing from fixed surveillance perspectives. Furthermore, most existing datasets are collected in developed countries with structured road infrastructure and do not represent dense, mixed-traffic environments found in developing cities.

To the best of our knowledge, there is no publicly available dataset specifically designed for illegal pedestrian road crossing detection in Dhaka or similar South Asian urban environments. Existing pedestrian-traffic research in Bangladesh has generally focused on behavioral analysis rather than producing fixed-camera surveillance datasets with pedestrian crossing violation annotations [17]. This creates a significant research gap, as models trained on existing benchmarks lack both the visual diversity and semantic labels required for reliable illegal crossing detection.

To address this gap, this work introduces a dedicated dataset collected from Dhaka urban environments with annotations representing different pedestrian-road interaction states. Unlike existing datasets, the proposed dataset explicitly considers illegal crossing behavior from fixed surveillance viewpoints and provides a foundation for developing real-time pedestrian safety monitoring systems.

3. Dataset

3.1. Data Collection and Dataset Diversity

The dataset was collected using four different smartphone cameras: Samsung A13, Samsung A55, Huawei P21, and Redmi 12. Using multiple devices helped minimize camera-specific bias and improved the model’s ability to generalize across images captured with different sensors and image processing techniques. More than 1,000 images were captured from several locations in Dhaka, including the University of Dhaka campus, Elephant Road, Bakshibazar, Motijheel, Gulistan, and the Dhaka–Chittagong Highway. The images were collected under a wide range of real-world conditions, with variations in lighting, camera angles, object-to-camera distance, and background complexity. This diversity was intentionally introduced to ensure that the trained model would perform reliably in different environments.

After data collection, all images were manually reviewed to remove duplicates, severely blurred images, and other redundant samples. Following this filtering process, a total of 693 high-quality images were retained for model training and evaluation. Figure 1a shows representative examples from the dataset, illustrating the diversity in capture conditions and scene characteristics.

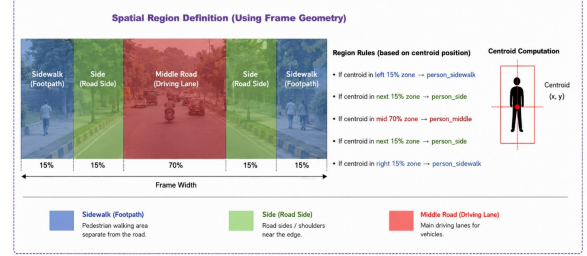
3.2. Annotation Strategy and Augmentations

The proposed region of a road is divided into five vertical zones: 15% sidewalk, 15% road-side, 70% middle road, 15% road-side, and 15% sidewalk. A detected pedestrian is assigned to `person_sidewalk`, `person_side`, or `person_middle` according to the horizontal position of the bounding-box centroid within these zones (Figure 1b).

All images were manually annotated with axis-aligned bounding boxes using Roboflow. Two



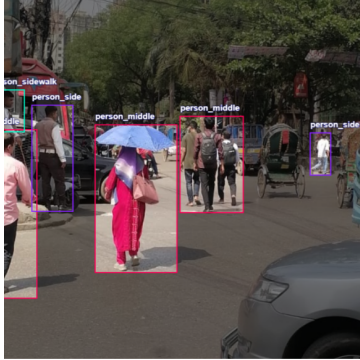
(a) Representative images illustrating diverse capture conditions.



(b) Definition of the pedestrian road-position regions.

Figure 1: Dataset examples and the spatial-region definition used for annotation.

annotation schemes were created from the same image set. In the multi-class scheme, pedestrians were labeled as *person_middle*, *person_side*, or *person_sidewalk* according to their position on the road, as shown in Figure 2(a). In the single-class scheme, all pedestrians in the middle position were labeled as a single class, *person_middle*, as shown in Figure 2(b).



(a) Multi-class annotation



(b) Single-class annotation

Figure 2: Examples of the proposed multi-class and single-class annotation schemes.

Data augmentation was applied during YOLO training. Mosaic augmentation was enabled for every training image, while MixUp and horizontal flipping were applied with probabilities of 0.1 and 0.5, respectively. Random geometric transformations included translation of up to 0.1 of the image dimensions and scaling with a gain of 0.5. Color augmentation was performed in the HSV color space using hue, saturation, and value gains of 0.015, 0.7, and 0.4, respectively. These transformations increased the diversity of the training samples and improved robustness to variations in pedestrian position, scale, orientation, lighting, and scene appearance.

3.3. Dataset Statistics

Table 1 summarizes the statistics of the two dataset variants. Dataset 1 contains three classes: *person_middle*, *person_side*, and *person_sidewalk*, whereas Dataset 2 contains a single class: *person_middle*.

The per-class annotation distributions for both datasets, separated by training, validation, and

Table 1: Summary statistics of the two datasets.

Property	Dataset 1 (Multi.)	Dataset 2 (Single)
Total images	682	671
Background-only images	28	207
Number of classes	3	1

Table 2: Class-wise annotation distribution across data splits for both datasets.

Dataset	Class	Train	Val	Test	Total
Dataset 1	person_middle	864	67	116	1047
	person_side	696	85	87	868
	person_sidewalk	877	163	90	1130
Dataset 2	person_middle	659	71	86	816

testing splits, are presented in Table 2. Dataset 1 demonstrates a relatively balanced distribution among its three pedestrian categories. The *person_sidewalk* category contains the highest number of annotations (1,130 instances), followed by *person_middle* (1,047 instances) and *person_side* (868 instances). The difference between the most frequent and least frequent classes is relatively small, indicating that the dataset does not suffer from severe class imbalance. This balanced distribution was achieved through data collection from diverse environments containing roadway, curbside, and sidewalk pedestrian activities. For Dataset 2, the single *person_middle* (person) class contains 659, 71, and 86 annotated instances in the training, validation, and testing splits, respectively.

4. Methodology

The proposed system is designed to detect illegal road crossing using a deep learning pipeline optimized for real-time surveillance video analysis. The framework consists of a single-stage object detection model that identifies pedestrians and classifies their spatial position into three categories: *person_sidewalk*, *person_side*, and *person_middle*.

4.1. System Overview

Given an input surveillance video, frames are extracted and passed through a YOLO-based object detection model. The model performs two tasks simultaneously: (1) detecting pedestrians and (2) assigning each detected pedestrian to a spatial class based on their location in the scene.

4.2. Detection and Classification Pipeline

The YOLO-based detector outputs bounding boxes in the form:

$$B = (x, y, w, h, c)$$

where (x, y) represents the center coordinates, (w, h) represent width and height, and c is the confidence score.

For each detected pedestrian, the centroid position is computed and mapped to one of the three spatial classes using predefined boundary rules based on the road layout in the dataset.

4.3. Decision Logic for Illegal Crossing

After classification, pedestrian behavior is determined as follows:

- **Legal behavior:** Pedestrians classified as *person_sidewalk*.
- **Risky behavior:** Pedestrians classified as *person_side*.
- **Illegal crossing:** Pedestrians classified as *person_middle*.

This rule-based interpretation allows the system to directly convert classification into actionable traffic safety signals.

5. Experiments

5.1. Experimental Setup and Training Configuration

All experiments were conducted in Google Colaboratory using NVIDIA Tesla T4 GPUs and a consistent Ultralytics YOLO pipeline. YOLOv8m, YOLOv8l, YOLO11m, and YOLO11l were trained separately under the common configuration summarized in Table 3. The best validation checkpoint (`best.pt`) from each run was used for final test-set evaluation.

Table 3: Training configuration used for the evaluated YOLO models.

Parameter	Value or description
Base models	YOLOv8m, YOLOv8l, YOLO11m, and YOLO11l
Tasks	Single-class and multi-class pedestrian detection
Input image size	960×960 pixels
Training epochs	50
Batch size	8
Optimizer	AdamW
Initial learning rate (lr_0)	5×10^{-4}
Learning-rate schedule	Cosine annealing
Weight decay	5×10^{-4}
Dropout	0.05
Label smoothing	0.05
Early-stopping patience	10 epochs
Hardware	NVIDIA Tesla T4 GPU (14,913 MiB VRAM)

5.2. Evaluation Metrics

Detector performance was evaluated using precision, recall, F1-score, Intersection over Union (IoU), mean Average Precision at an IoU threshold of 0.5 (mAP@0.5), and mean Average Precision averaged over IoU thresholds from 0.5 to 0.95 (mAP@0.5:0.95). Together, these metrics measure prediction reliability, detection completeness, the balance between precision and recall, and bounding-box localization quality.

Let TP , FP , and FN denote the numbers of true-positive, false-positive, and false-negative detections, respectively. Precision is the proportion of predicted detections that are correct:

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (1)$$

Recall is the proportion of ground-truth objects correctly detected:

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (2)$$

Precision penalizes false alarms, whereas recall penalizes missed pedestrians. Because both error types are important in road-safety monitoring, their harmonic mean is reported as the F1-score:

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3)$$

Table 4: Validation and test performance on the single-class dataset.

Model	Precision		Recall		F1-score		mAP@0.5		mAP@0.5:0.95	
	Val.	Test	Val.	Test	Val.	Test	Val.	Test	Val.	Test
YOLOv8m	0.8528	0.7748	0.8310	0.7600	0.8418	0.7673	0.9036	0.8063	0.7354	0.6271
YOLOv8l	0.7461	0.6155	0.7866	0.8023	0.7658	0.6966	0.8519	0.7573	0.6856	0.5662
YOLO11m	0.7564	0.5802	0.7042	0.7232	0.7294	0.6439	0.8333	0.7113	0.6056	0.5224
YOLO11l	0.7331	0.7604	0.8310	0.8117	0.7790	0.7852	0.8699	0.8419	0.6763	0.6394

IoU measures the spatial agreement between a predicted bounding box B_p and its corresponding ground-truth box B_{gt} :

$$\text{IoU}(B_p, B_{gt}) = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|}. \quad (4)$$

At a specified IoU threshold, Average Precision (AP) summarizes the precision–recall curve for one class. Conceptually, it is expressed as

$$\text{AP} = \int_0^1 p(r) dr, \quad (5)$$

where $p(r)$ denotes interpolated precision as a function of recall r . For C classes, mean Average Precision is the arithmetic mean of the class-wise AP values:

$$\text{mAP} = \frac{1}{C} \sum_{c=1}^C \text{AP}_c. \quad (6)$$

The mAP@0.5 metric uses a single IoU threshold of 0.5 and therefore provides a relatively tolerant assessment of localization. The stricter mAP@0.5:0.95 metric averages mAP over ten thresholds from 0.50 to 0.95 in increments of 0.05:

$$\text{mAP}_{0.50:0.95} = \frac{1}{10} \sum_{k=0}^9 \text{mAP}_{\text{IoU}=0.50+0.05k}. \quad (7)$$

Accordingly, mAP@0.5 primarily indicates whether pedestrians are detected with reasonable overlap, whereas mAP@0.5:0.95 is more sensitive to precise bounding-box placement. Reporting both metrics provides a complementary assessment of detection and localization performance.

6. Results and Discussion

6.1. Single-Class Detection Results

Table 4 compares the four YOLO variants on the single-class pedestrian-detection task. On the independent test set, YOLO11l achieved the highest recall (0.8117), F1-score (0.7852), mAP@0.5 (0.8419), and mAP@0.5:0.95 (0.6394). These results indicate that YOLO11l provided the strongest overall balance between pedestrian recovery and localization accuracy. YOLOv8m obtained the highest test precision (0.7748), but its recall and mAP values were lower than those of YOLO11l. Thus, YOLOv8m produced slightly fewer false-positive detections, whereas YOLO11l detected more pedestrians and localized them more accurately across stricter IoU thresholds.

Table 5: Validation and test performance on the multi-class dataset.

Model	Precision		Recall		F1-score		mAP@0.5		mAP@0.5:0.95	
	Val.	Test	Val.	Test	Val.	Test	Val.	Test	Val.	Test
YOLOv8m	0.9845	0.8135	0.8889	0.8519	0.9343	0.8323	0.9612	0.8377	0.6734	0.5004
YOLOv8l	0.9777	0.8761	0.8889	0.5185	0.9312	0.6515	0.9469	0.6767	0.6788	0.3663
YOLO11m	0.9871	0.8669	0.6667	0.8148	0.7959	0.8400	0.7811	0.8547	0.6191	0.4825
YOLO11l	0.9845	0.8629	0.6667	0.5556	0.7950	0.6760	0.7618	0.6895	0.6025	0.4219

The validation-to-test differences also reveal differences in generalization. YOLO11l exhibited the smallest decrease in mAP@0.5:0.95, falling by only 0.0369 from validation to test. In comparison, the decreases for YOLOv8m, YOLOv8l, and YOLO11m were 0.1083, 0.1194, and 0.0832, respectively. Therefore, the test results suggest that YOLO11l was not only the most accurate single-class model but also the most stable under the distribution shift between the validation and test sets. YOLOv8l achieved competitive recall but its comparatively low precision (0.6155) indicates that this sensitivity was accompanied by a larger number of false-positive detections. YOLO11m produced the lowest test mAP values and was therefore the weakest model for this task.

6.2. Multi-Class Detection Results

Table 5 presents the results for the multi-class task. YOLOv8m achieved the highest test recall (0.8519) and the highest mAP@0.5:0.95 (0.5004), demonstrating the best overall performance under the stricter localization criterion. YOLO11m achieved the highest F1-score (0.8400) and mAP@0.5 (0.8547), while YOLOv8l achieved the highest precision (0.8761). However, both YOLOv8l and YOLO11l produced substantially lower recall values of 0.5185 and 0.5556, respectively. This indicates that their high precision was achieved by making more conservative predictions and missing a larger proportion of true pedestrian instances.

Although YOLO11m slightly outperformed YOLOv8m at $\text{IoU} = 0.5$, YOLOv8m was better when localization was evaluated over the full range of IoU thresholds. Its test mAP@0.5:0.95 exceeded that of YOLO11m by 0.0179, and its recall was higher by 0.0371. Consequently, YOLOv8m is the preferable multi-class model when missed detections and accurate bounding-box placement are both important. The large validation-to-test reductions observed for YOLOv8l and YOLO11l further suggest weaker generalization, particularly for recall and strict-IoU localization.

6.3. Comparative Analysis Across Models and Datasets

The results demonstrate that increasing model size did not consistently improve performance. For the single-class task, the large YOLO11l model clearly outperformed its medium counterpart, improving test mAP@0.5:0.95 from 0.5224 to 0.6394. In contrast, the medium YOLOv8m model was substantially better than YOLOv8l on the multi-class task, achieving a test mAP@0.5:0.95 of 0.5004 compared with 0.3663. A similar pattern occurred within the YOLO11 family on the multi-class dataset, where YOLO11m outperformed YOLO11l in recall, mAP@0.5, and mAP@0.5:0.95. These findings show that additional capacity alone does not guarantee better generalization; the interaction among architecture, model scale, label complexity, and dataset size is also important.

The two tasks also exhibited different performance characteristics. The best single-class model achieved a test mAP@0.5:0.95 of 0.6394, whereas the best multi-class model achieved 0.5004. The 0.1390 difference indicates that precise localization and classification were more difficult when pedestrians had to be separated into multiple road-position categories. This is expected because the multi-class labels encode contextual position rather than visually distinct object types. Nevertheless, the best multi-class model achieved a higher recall (0.8519) than the best

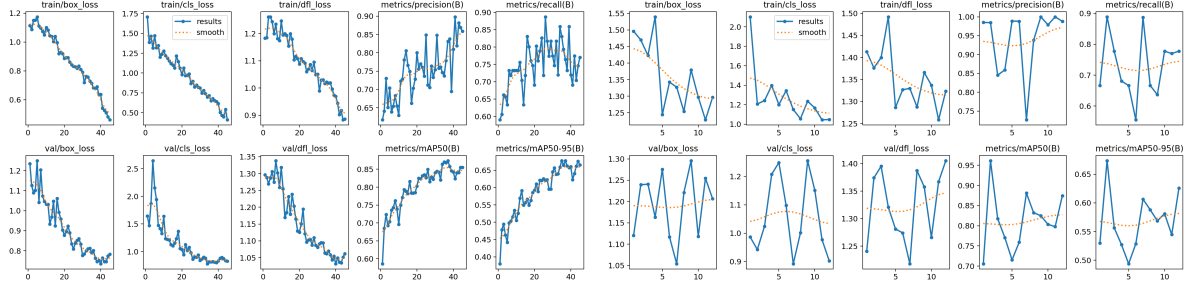


Figure 3: Training and validation dynamics of the selected baseline models: YOLO11l for the single-class task and YOLOv8m for the multi-class task.

single-class model (0.8117), showing that the principal difficulty was not pedestrian discovery alone, but the combination of localization and correct positional classification.

Model behavior also varied between the two datasets. For example, YOLO11l was the strongest single-class model but performed relatively poorly on the multi-class task, where its test recall fell to 0.5556 and its mAP@0.5:0.95 fell to 0.4219. Conversely, YOLOv8m remained competitive across both tasks and was the most reliable multi-class model. Therefore, the results do not support selecting one architecture solely on the basis of its performance on another labeling scheme. Model selection should instead be task-specific and should emphasize independent test-set performance.

6.4. Discussion

Overall, the experiments show a clear precision–recall trade-off. Models such as YOLOv8l on the multi-class dataset achieved high precision but low recall, indicating conservative detection behavior. In a road-safety application, this trade-off is important because missed pedestrians may be more consequential than occasional false alarms. On this basis, YOLOv8m is more suitable for the multi-class illegal-crossing task because it recovered the largest proportion of pedestrians while retaining the best strict-IoU mAP. For single-class pedestrian detection, YOLO11l offered the strongest balance, with the highest recall and both mAP measures on the test set.

The substantial gaps between validation and test performance for several models suggest that the validation split did not fully capture the diversity of the independent test set. Differences in scene density, camera viewpoint, illumination, pedestrian scale, and occlusion may have contributed to these gaps. The unusually higher test mAP@0.5 of YOLO11m on the multi-class task compared with its validation value also indicates that the two splits may differ in difficulty. These observations reinforce the importance of reporting independent test results rather than selecting models solely from validation performance.

6.5. Limitations

The comparison is limited to four YOLO variants and does not isolate every architectural difference between the YOLOv8 and YOLO11 families. In addition, parameter count, inference latency, and computational cost were not included with the supplied results, so the present comparison focuses on predictive performance rather than the complete accuracy–efficiency trade-off. The validation and test sets are also relatively small, and their apparent distribution differences may increase metric variance. Finally, class-wise AP and confusion matrices were not provided for the multi-class task; therefore, the analysis cannot determine which positional category contributed most strongly to the remaining errors.

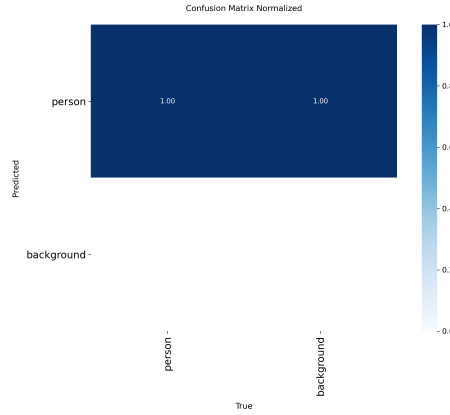


Figure 4: Confusion matrix normalized for YOLOv8m(multiclass)

6.6. Training Dynamics and Error Analysis

6.6.1 Training Convergence

Figure 3 presents the training and validation dynamics of the two models selected as baselines for the ablation study. The loss curves generally decreased during training, while the detection metrics improved and gradually stabilized. This indicates that both models were able to learn meaningful representations from the training data.

The curves also provide information that cannot be obtained from the final evaluation metrics alone. In particular, the relationship between training and validation performance provides an indication of the models' ability to generalize beyond the training samples. The YOLO11l model achieved strong single-class test performance, with a test $mAP@0.5:0.95$ of 0.6394, while YOLOv8m achieved the strongest multi-class performance under the same metric, with a value of 0.5004. Thus, the training curves provide a complementary view of the optimization process, whereas the independent test results remain the primary basis for model selection.

6.6.2 Detection Behavior and Error Analysis

Figure 3 presents the training and validation. The normalized confusion matrix of the selected multi-class model is shown in Figure 4. The matrix provides a more detailed view of the errors underlying the aggregate detection metrics. In particular, confusions between adjacent road-position categories are expected because the class definitions are spatially related rather than based on visually distinct object appearances. A pedestrian near the boundary between the roadside and the middle of the road may therefore exhibit visual characteristics associated with more than one category.

This observation is important because the multi-class task requires the model to perform both pedestrian detection and spatial-position classification. Consequently, an error may occur even when a pedestrian is correctly localized but assigned to the wrong positional category. The confusion matrix therefore complements the mAP results by revealing the types of classification errors that contribute to the overall performance.

6.6.3 Qualitative Detection Examples

Figure 5 compares the ground-truth annotations with the corresponding model predictions for the selected single-class detectors.



(a) YOLO11l single-class results: ground-truth annotations (left) and corresponding model predictions (right).



(b) YOLOv8m single-class results: ground-truth annotations (left) and corresponding model predictions (right).

Figure 5: Qualitative comparison of ground-truth annotations and model predictions for the selected single-class detectors.

7. Ablation Study

7.1. Baseline Model Selection

The best-performing model for each task was selected as the ablation baseline using independent test-set performance. Because $mAP@0.5:0.95$ evaluates detection across both lenient and strict IoU thresholds, it was treated as the primary selection metric, with precision and recall used as supporting criteria. YOLO11l was selected for the single-class task because it achieved the highest test $mAP@0.5:0.95$ (0.6394), together with the highest recall and $mAP@0.5$. YOLOv8m was selected for the multi-class task because it achieved the highest test $mAP@0.5:0.95$ (0.5004)

Table 6: Input-resolution ablation for the selected single-class and multi-class baseline models.

Task	Model	Size	Validation					Test				
			P	R	F1	mAP50	mAP50-95	P	R	F1	mAP50	mAP50-95
Single-class	YOLO11l	640	0.8028	0.7452	0.7729	0.8118	0.6436	0.7393	0.8023	0.7695	0.7803	0.5798
Single-class	YOLO11l	960	0.7331	0.8310	0.7790	0.8699	0.6763	0.7604	0.8117	0.7852	0.8419	0.6394
Single-class	YOLO11l	1280	0.9227	0.8889	0.9055	0.9080	0.6050	0.8327	0.7407	0.7840	0.8049	0.4932
Multi-class	YOLOv8m	640	0.9866	0.6667	0.7957	0.7628	0.5939	0.9500	0.7036	0.8084	0.8014	0.4885
Multi-class	YOLOv8m	960	0.9845	0.8889	0.9343	0.9612	0.6734	0.8135	0.8519	0.8323	0.8377	0.5004
Multi-class	YOLOv8m	1280	0.7564	0.7042	0.7294	0.8333	0.6056	0.5802	0.7232	0.6439	0.7113	0.5224

and recall (0.8519), while maintaining competitive precision and mAP@0.5.

The comparison among YOLOv8m, YOLOv8l, YOLO11m, and YOLO11l can itself be interpreted as a model-variant ablation. It shows that the preferred scale depends on the task: the large YOLO11 model was advantageous for single-class detection, whereas the medium YOLOv8 model generalized better for multi-class positional detection. This result suggests that the additional capacity of a large model is useful only when it matches the complexity and amount of available training data.

7.2. Input-Resolution Ablation

The effect of input resolution was evaluated at 640, 960, and 1280 pixels using the selected baseline for each task: YOLO11l for single-class detection and YOLOv8m for multi-class detection. All other available training and evaluation settings were kept unchanged so that the observed differences could be attributed primarily to image resolution. Table 6 summarizes the validation and test results.

For the single-class task, increasing the resolution from 640 to 960 pixels improved every test metric. Precision increased from 0.7393 to 0.7604, recall from 0.8023 to 0.8117, F1-score from 0.7695 to 0.7852, mAP@0.5 from 0.7803 to 0.8419, and mAP@0.5:0.95 from 0.5798 to 0.6394. Thus, 960 pixels provided the best overall single-class configuration. Increasing the resolution further to 1280 pixels raised precision to 0.8327 but reduced recall to 0.7407 and mAP@0.5:0.95 to 0.4932. Its F1-score remained competitive at 0.7840 because the gain in precision largely offset the recall reduction; however, the strict-IoU mAP shows that localization quality deteriorated. The decline may reflect increased optimization difficulty or overfitting to high-resolution details that did not transfer to the test set.

The multi-class task exhibited a different trade-off. At 640 pixels, YOLOv8m achieved the highest test precision (0.9500) but a comparatively low recall of 0.7036, resulting in an F1-score of 0.8084. Increasing the resolution to 960 pixels improved recall to 0.8519 and produced the highest F1-score (0.8323) and mAP@0.5 (0.8377), making it the most balanced operating point. At 1280 pixels, mAP@0.5:0.95 increased slightly to 0.5224, but precision fell to 0.5802 and the F1-score dropped substantially to 0.6439. Therefore, 1280 pixels is preferable only when strict-IoU localization is the overriding selection criterion, whereas 960 pixels provides a better precision-recall balance and coarse-to-moderate localization accuracy.

Taken together, the ablation results demonstrate that detection performance does not increase monotonically with input resolution. The optimal resolution is task-dependent: 960 pixels is clearly preferred for single-class detection and offers the most balanced multi-class performance, while 1280 pixels provides only a modest multi-class gain in mAP@0.5:0.95 at the cost of substantial reductions in other metrics. Since higher resolutions also require more memory and inference time, the 960-pixel configuration represents the most practical default for the proposed system unless strict multi-class localization is prioritized above runtime efficiency and detection recall.

8. Conclusion

This work introduced a custom dataset for illegal pedestrian road-crossing detection in Dhaka, along with single-class and spatially aware multi-class annotation schemes, and evaluated four YOLO variants (YOLOv8m, YOLOv8l, YOLO11m, and YOLO11l) for this task. YOLO11l achieved the best single-class performance (test mAP@0.5:0.95 of 0.6394), while YOLOv8m was the most reliable multi-class model (test mAP@0.5:0.95 of 0.5004, recall of 0.8519). The results show that larger models do not consistently generalize better for spatially structured pedestrian classification, and that model selection should be task-specific rather than architecture-specific. Resolution ablation further identified 960×960 as the most practical operating point, balancing precision, recall, and localization accuracy across both tasks. Future work will extend the dataset with more locations and traffic conditions, incorporate class-wise error analysis for the multi-class task, and explore temporal or tracking-based approaches to improve robustness in dense, unstructured urban traffic environments.

Project Resources

The source code, notebooks, and dataset associated with this study are available at:

- **GitHub Repository:** <https://github.com/Tawyabulislamtamim/Illegal-Road-Crossing-Detection>
- **Dataset:** <https://drive.google.com/drive/folders/1v-Gq4tBK6j-LUvTfFIBTi2oS67OiXVxk?usp=sharing>

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